

INFOSYS SPRINGBOARD 7.0

VIRTUAL INTERNSHIP

INTELLIGENT COGNITIVE ALARM PLATFORM

Theory Documentation (Milestone-1)

Database Layer

Submitted By

Name: Abdul Raheem

Mentor: Sai Jyoteesh

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1. Project Description

The Intelligent Cognitive Alarm Platform is an AI-powered platform designed to help users develop consistent wake-up habits by requiring cognitive challenges before alarm dismissal. The platform analyses user behaviour, wake-up performance, snooze patterns, sleep schedules and cognitive engagement to encourage healthy routines.

My assigned responsibility in this internship is only the Database Layer. Therefore, this document explains only the database design and implementation completed using PostgreSQL and MongoDB.

Source: Project PDF.

2. Objective

The objective of the database layer is to provide reliable storage for structured and document-based data required by the Intelligent Cognitive Alarm Platform. PostgreSQL is used as the primary relational database and MongoDB is used as the secondary document database.

Implementation decision: The report documents only my database contribution.

3. Database Technologies Used

PostgreSQL (Primary Database)

- users
- alarms
- habits
- scores
- reports

MongoDB (Secondary Database)

- challenge_data
- user_preferences
- logs
- notifications
- content

4. Development Environment

Operating System: Windows

Database Tool: pgAdmin 4

NoSQL Tool: MongoDB Compass

Version Control: Git & GitHub

5. Database Architecture

The project uses PostgreSQL for structured relational data and MongoDB for flexible document-based data. The relational database stores core entities while MongoDB stores document collections supporting platform data.

This follows the project technology stack.

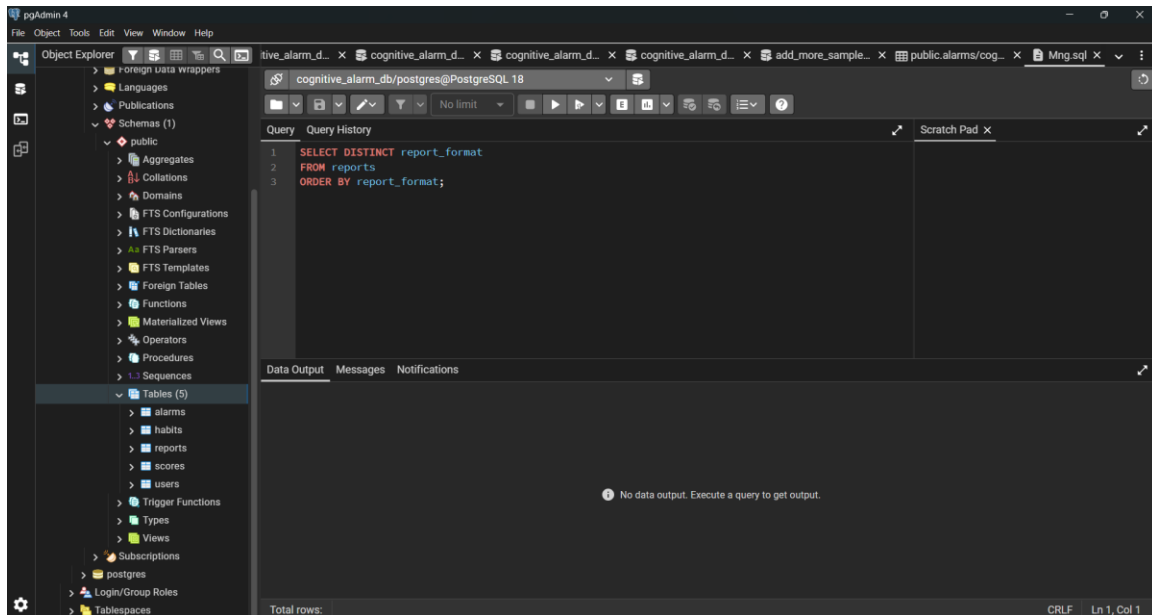
6. PostgreSQL Database Implementation

Completed:

- Database creation
- Table creation
- Primary Keys
- Foreign Keys
- Sample data insertion
- SQL validation

Tables:

users, alarms, habits, scores, reports



7. MongoDB Database Implementation

Database: cognitive_alarm_db

Collections:

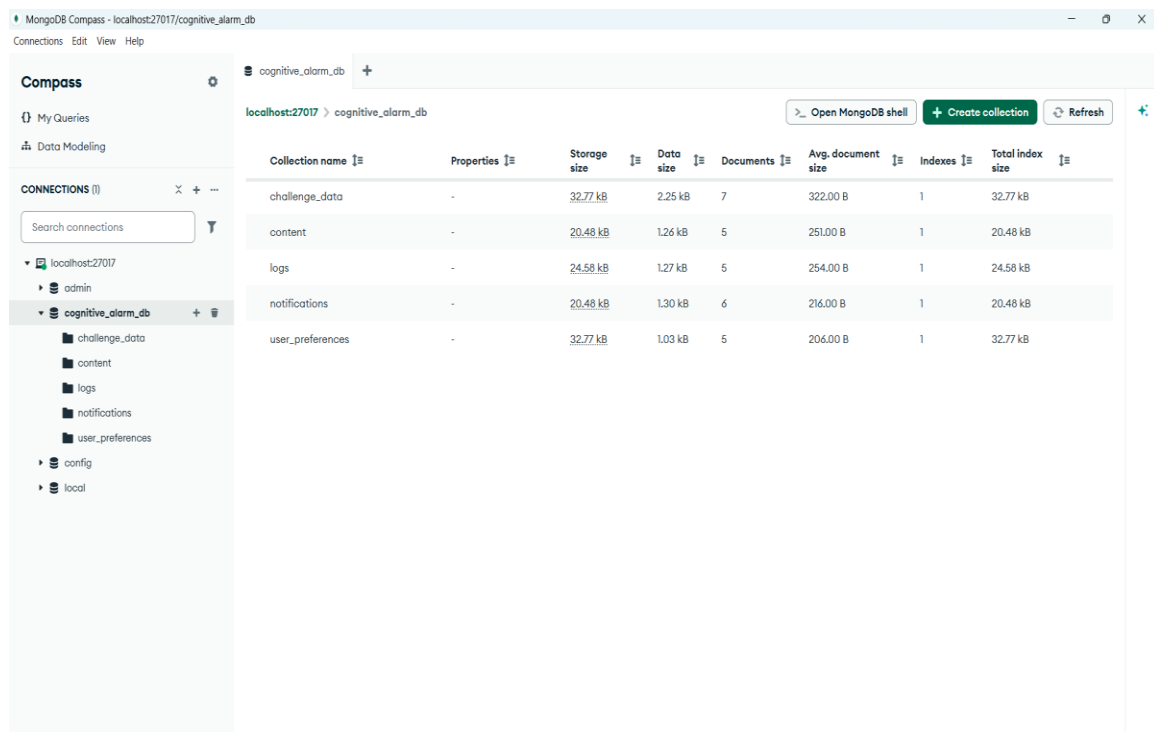
challenge_data

user_preferences

logs

notifications

content



MongoDB Compass - localhost27017/cognitive_alarm_db

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (1)

Search connections

localhost:27017

- admin
- cognitive_alarm_db**
 - challenge_data
 - content
 - logs
 - notifications
 - user_preferences
- config
- local

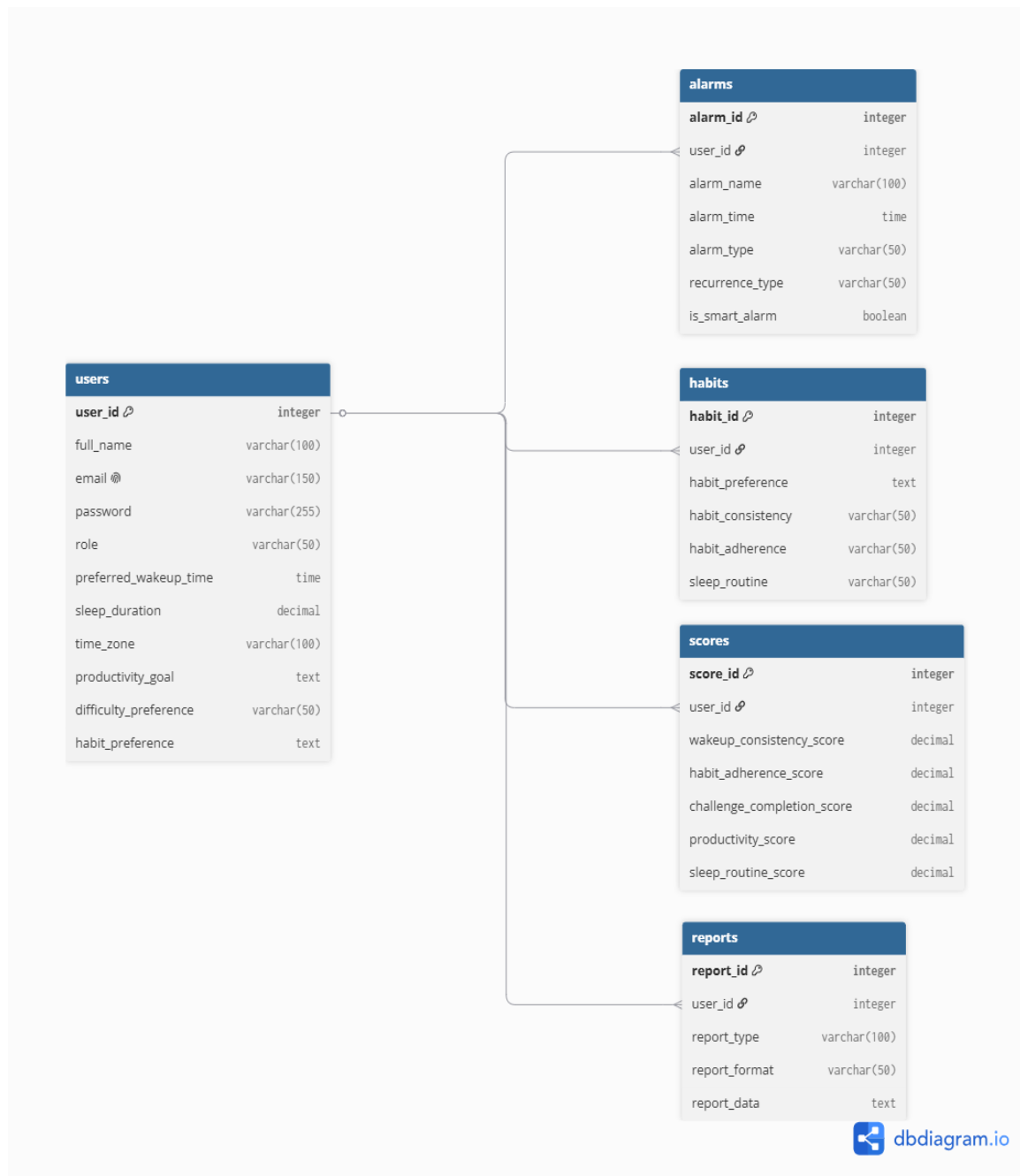
localhost:27017 > cognitive_alarm_db

> Open MongoDB shell + Create collection Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
challenge_data	-	32.77 kB	2.25 kB	7	322.00 B	1	32.77 kB
content	-	20.48 kB	1.26 kB	5	251.00 B	1	20.48 kB
logs	-	24.58 kB	1.27 kB	5	254.00 B	1	24.58 kB
notifications	-	20.48 kB	1.30 kB	6	216.00 B	1	20.48 kB
user_preferences	-	32.77 kB	1.03 kB	5	206.00 B	1	32.77 kB

8. Entity Relationship Diagram

The ER Diagram represents relationships among PostgreSQL tables.



9. Database Validation and Testing

Validation performed:

- Table creation verified
- Foreign keys verified
- Sample records inserted
- SQL queries executed successfully
- MongoDB document insertion verified

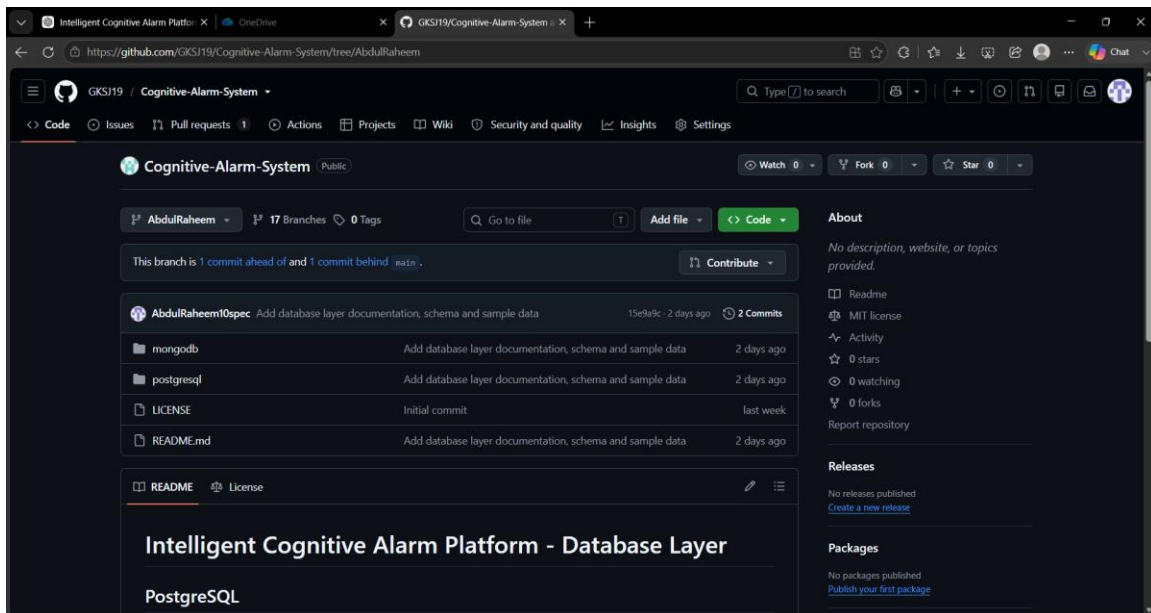
10. GitHub Repository

Repository contains:

postgresql/

mongodb/

README.md



11. Challenges Faced

- Designing relational relationships.
- Maintaining data consistency.
- Organizing relational and document databases.
- Preparing documentation.

12. Learning Outcomes

- PostgreSQL database design
- MongoDB collections
- ER Diagram design
- Primary and Foreign Keys
- Database validation
- GitHub version control
- Technical documentation

13. Conclusion

The Database Layer for the Intelligent Cognitive Alarm Platform was successfully implemented using PostgreSQL and MongoDB. The work includes relational schema design, document collections, validation, backup, and GitHub version control. This provides a reliable data foundation for the overall platform.

14. References

1. Intelligent Cognitive Alarm Platform Project PDF
2. PostgreSQL Documentation
3. MongoDB Documentation
4. Git Documentation